

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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1. TITLE

Develop Dynamic Webpage Using JavaScript

2. AIM

To develop and execute a dynamic webpage using JavaScript to manipulate webpage content, handle user interactions, and dynamically update elements in the web page.

3. OBJECTIVE

- To understand the role of JavaScript in developing dynamic web pages.
- To modify HTML content dynamically using JavaScript.
- To access and manipulate HTML elements using the DOM.
- To handle user events such as button clicks.
- To use JavaScript functions to perform specific operations.
- To dynamically change text and display information on a webpage.
- To execute and test the dynamic webpage in a web browser.

4. THEORY

JavaScript is a scripting language used to make web pages interactive and dynamic. Unlike static HTML pages, JavaScript can change webpage content and behavior while the page is running.

The Document Object Model (DOM) represents an HTML document as a collection of objects. JavaScript can access these objects and modify their content, attributes, and properties dynamically.

JavaScript can select HTML elements using methods such as `getElementById()`. After selecting an element, its content can be changed using properties such as `innerHTML` or `textContent`.

JavaScript functions are reusable blocks of code that perform specific tasks. Event handling allows functions to execute when users interact with webpage elements, such as clicking a button.

In this experiment, a webpage is created with HTML elements and JavaScript is used to dynamically change the displayed message when the user interacts with the button.

5. ALGORITHM / PROCEDURE

1. Start the program.
2. Create an HTML file containing the basic webpage structure.

3. Add a heading, input field, paragraph, and button to the webpage.
4. Create a JavaScript function to perform the dynamic operation.
5. Access the required HTML element using its ID.
6. Read the value entered by the user.
7. Attach the function to the button click event.
8. Change the webpage content dynamically using JavaScript.
9. Save the file with an appropriate .html extension.
10. Open the HTML file in a web browser.
11. Enter a value and click the button.
12. Verify that the webpage content changes dynamically.
13. Test the webpage with different input values.
14. Stop the program.

6. PROGRAM

```
<!DOCTYPE html>
<html>
<head>
    <title>Dynamic Webpage Using JavaScript</title>
</head>
<body>

    <h2>Dynamic Webpage</h2>

    <input type="text" id="name" placeholder="Enter your name">
    <button onclick="displayMessage()">Click Me</button>

    <p id="message">Your message will appear here.</p>

    <script>
        function displayMessage() {
            let name = document.getElementById("name").value;

            if (name === "") {
                document.getElementById("message").innerHTML =
                    "Please enter your name.";
            } else {
                document.getElementById("message").innerHTML =
                    "Welcome, " + name + "!";
            }
        }
    </script>

</body>
</html>
```

7. CODE EXPLANATION

Tag / Element / Concept	Purpose
<input>	Accepts the user's name as input.
<button>	Provides a button to trigger the JavaScript function.
onclick	Calls displayMessage() when the button is clicked.
function displayMessage()	Defines the JavaScript function for the dynamic operation.
getElementById()	Selects an HTML element using its ID.
value	Gets the value entered in the input field.
if...else	Checks whether the user has entered a name.
innerHTML	Changes paragraph content dynamically.
document	Represents the current HTML document and provides access to its elements.

8. TEST CASES AND EXECUTION DATA

Test Case	Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a browser.	Dynamic webpage loads successfully.	Webpage loaded correctly.	Pass
TC-02	Enter the name 'Mohini' and click the button.	Welcome message should be displayed.	Welcome, Mohini! displayed.	Pass
TC-03	Leave the input field empty and click the button.	Validation message should be displayed.	Please enter your name. displayed.	Pass
TC-04	Enter another name and click the button.	Message should update with the new name.	Message updated correctly.	Pass
TC-05	Click the button with different inputs.	Webpage content should change dynamically each time.	Content changed correctly.	Pass

9. OUTPUT

Output 1: Dynamic Webpage before user interaction.

```
<!DOCTYPE html>
<html>
<head>
  <title>Dynamic Webpage Using JavaScript</title>
</head>
<body>

  <h2>Dynamic Webpage</h2>

  <input type="text" id="name" placeholder="Enter your name">
  <button onclick="displayMessage()">Click Me</button>

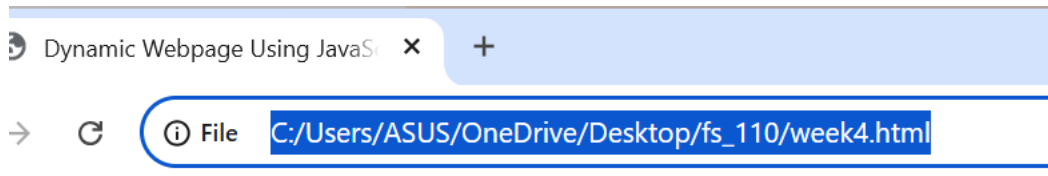
  <p id="message">Your message will appear here.</p>

  <script>
    function displayMessage() {
      let name = document.getElementById("name").value;

      if (name === "") {
        document.getElementById("message").innerHTML =
          "Please enter your name.";
      } else {
        document.getElementById("message").innerHTML =
          "Welcome, " + name + "!";
      }
    }
  </script>

</body>
</html>
```

Output 1:



Output 2:

Dynamic Webpage

Your message will appear here.

Output 3

Dynamic Webpage

Welcome, mohini!

Output 4: Screenshot of the executed dynamic webpage.

10. RESULT

Thus, a dynamic webpage was successfully developed and executed using JavaScript. The program demonstrated DOM manipulation, event handling, functions, user input, and dynamic modification of webpage content. The webpage was tested with different inputs, and the expected results were obtained successfully.